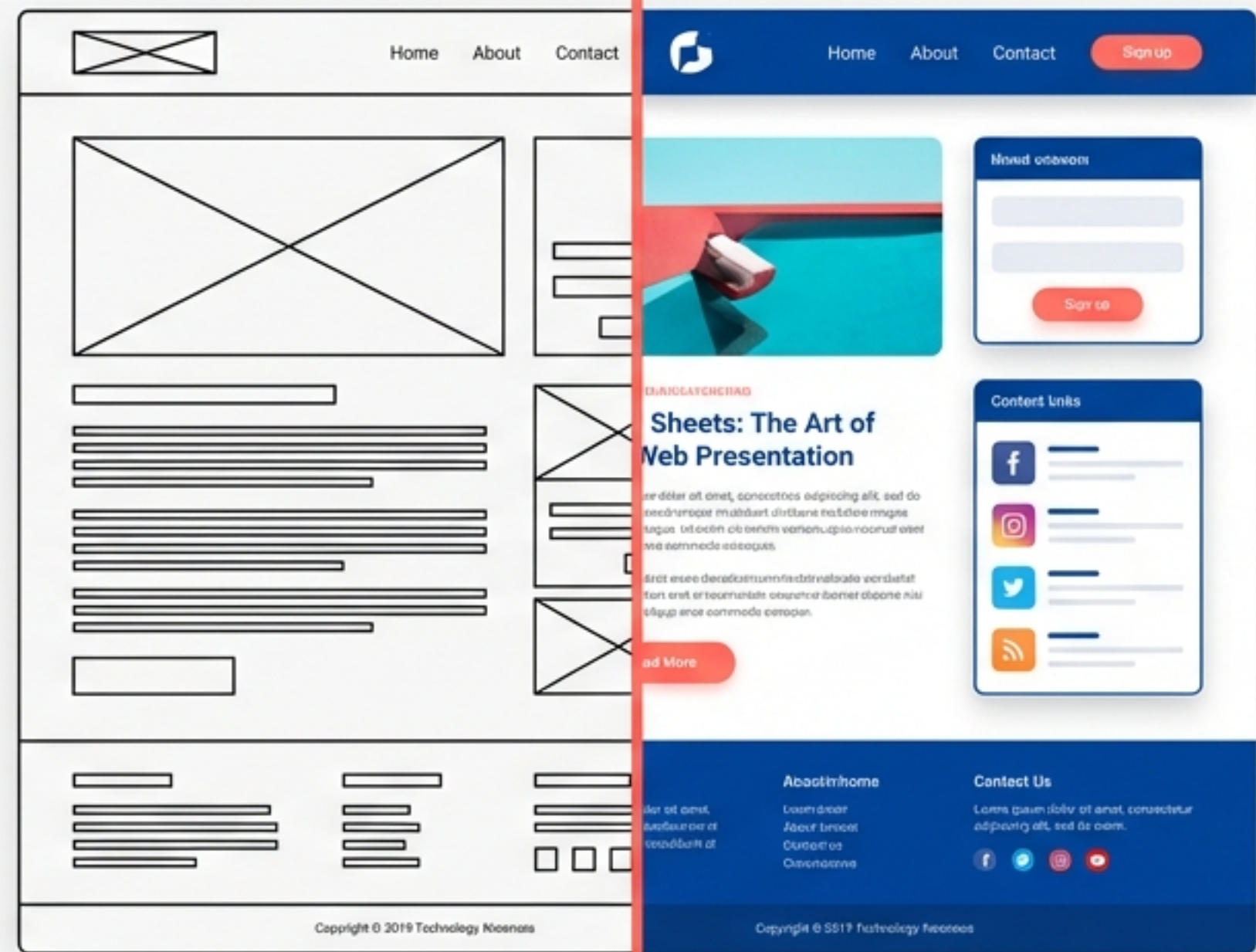


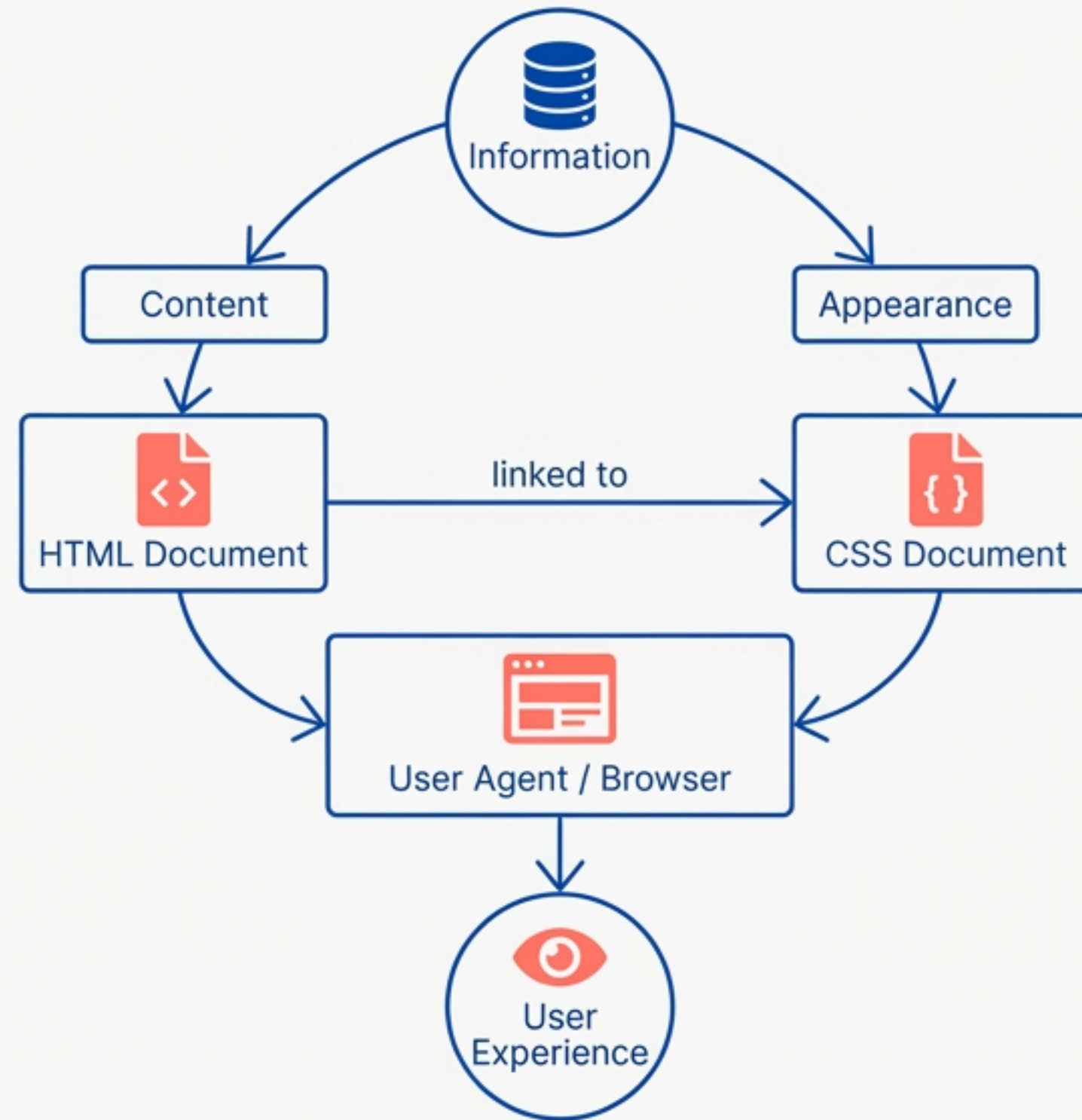
Cascading Style Sheets: The Art of Web Presentation

Fundamentals of Design, Syntax, and Implementation



THE SISTER TECHNOLOGY TO HTML

The Separation of Concerns



CSS allows us to decouple the structure of a document from its visual presentation.
This separation is the foundation of modern web development.

Why CSS Matters: Efficiency & Performance



Global Control

Change the look of an entire multi-page site by editing a single file.



Performance

Pages load faster due to cached style sheets and less code duplication.



Workflow

Decreased production work saves time during development.



Precision

Rich design options (layout, typography, color) that HTML alone cannot provide.

The Evolution of Style

CSS1

The Basics

Fonts, Colors, Alignment, Spacing

This is a basic text block
with simple formatting.

CSS2 / 2.1

The Structure

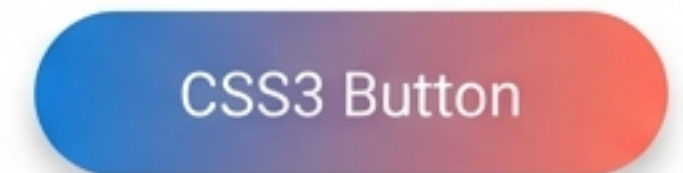
Layouts and Positioning (CSS-P)



CSS3

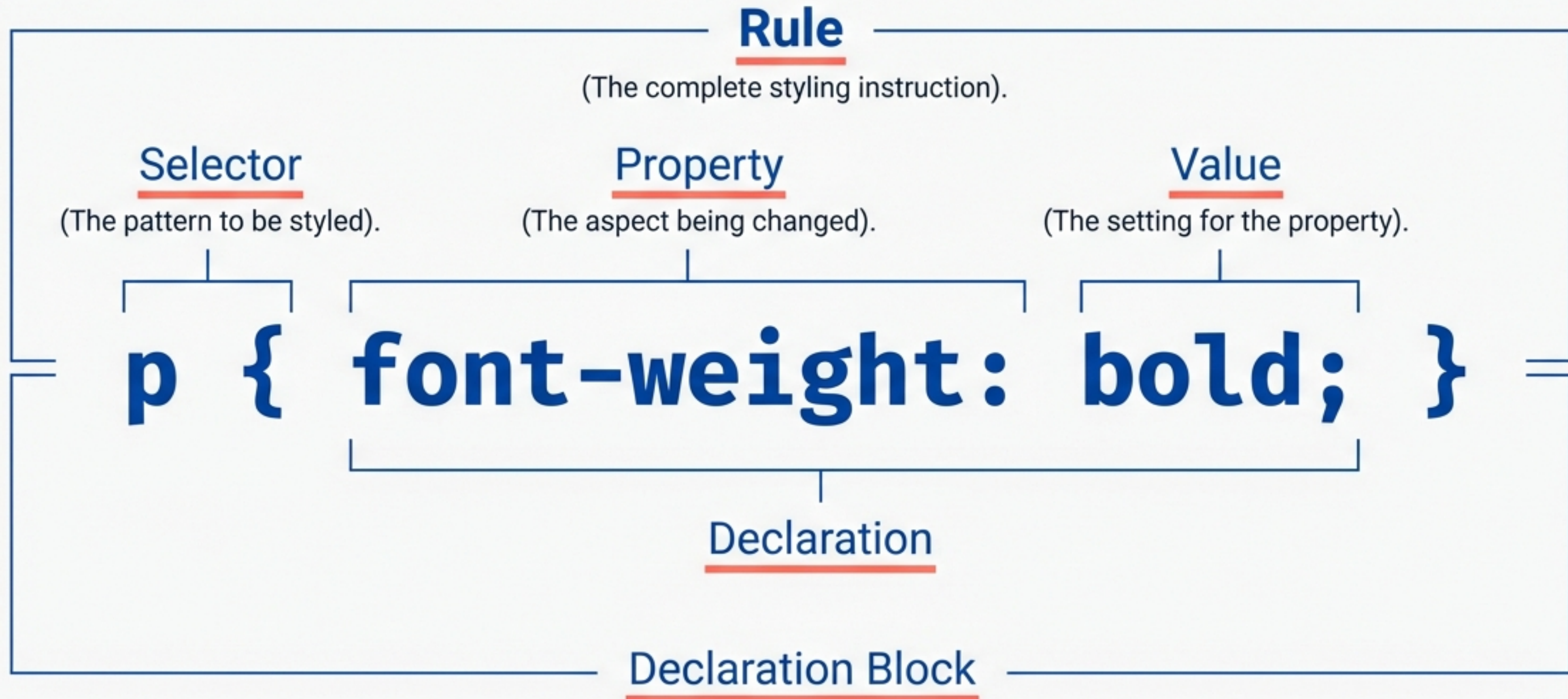
The Modern Era

Visual Effects and Advanced Sizing



"We have moved from simple text formatting to complex, interactive layouts."

The Anatomy of a Rule



Grouping & Efficiency

DRY: Don't Repeat Yourself

Inefficient

```
h1 {  
  font-family: "sans-serif";  
}
```

```
h2 {  
  font-family: "sans-serif";  
}
```

```
h3 {  
  font-family: "sans-serif";  
}
```

Efficient

```
h1, h2, h3 {  
  font-family: "sans-serif";  
}
```

Group selectors by separating them with commas to apply the same properties to multiple elements without repetition.

Connecting the Sheets

Three methods to apply CSS to HTML



Embedding

Internal styles placed in the <head> section using <style> tags.



Linking

Best Practice

Connecting to an external .css file using the <link> tag.



Inline

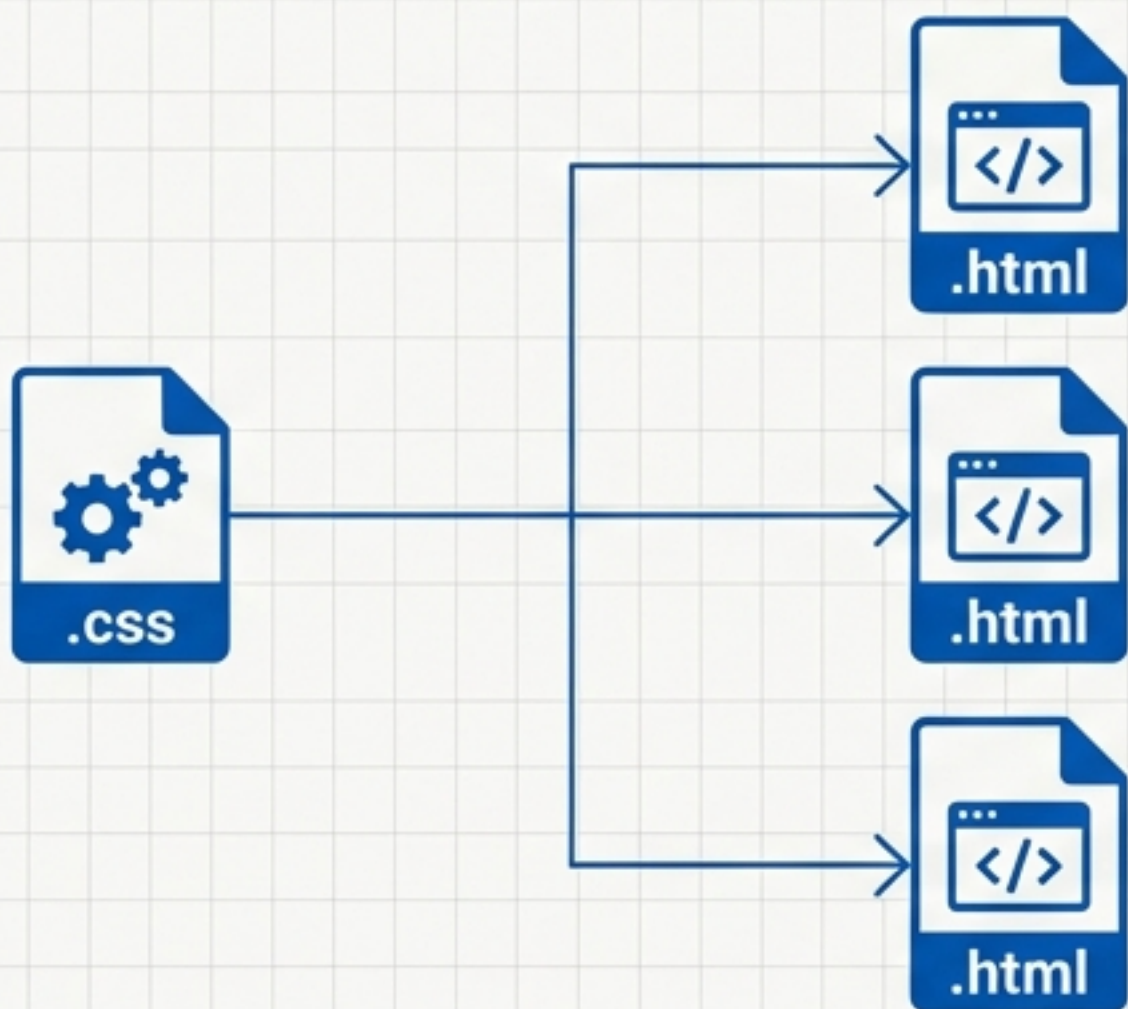
Styles plunked directly into HTML tags.

Avoid

Method 1: The External Style Sheet

Highly Recommended

Ideal for applying style to many pages. You can change the look of an entire website by changing just one file.



HTML <head>

```
<head>
  <link rel="stylesheet"
type="text/css" href="style.css">
</head>
```

Alternative import syntax: @import "style1.css";

Methods 2 & 3: Exceptions & Warnings

Internal (Embedded)

Defined in the <head> using <style>. Use only when a single document requires a unique style.



```
<style>
  h1 { color: blue; }
</style>
```



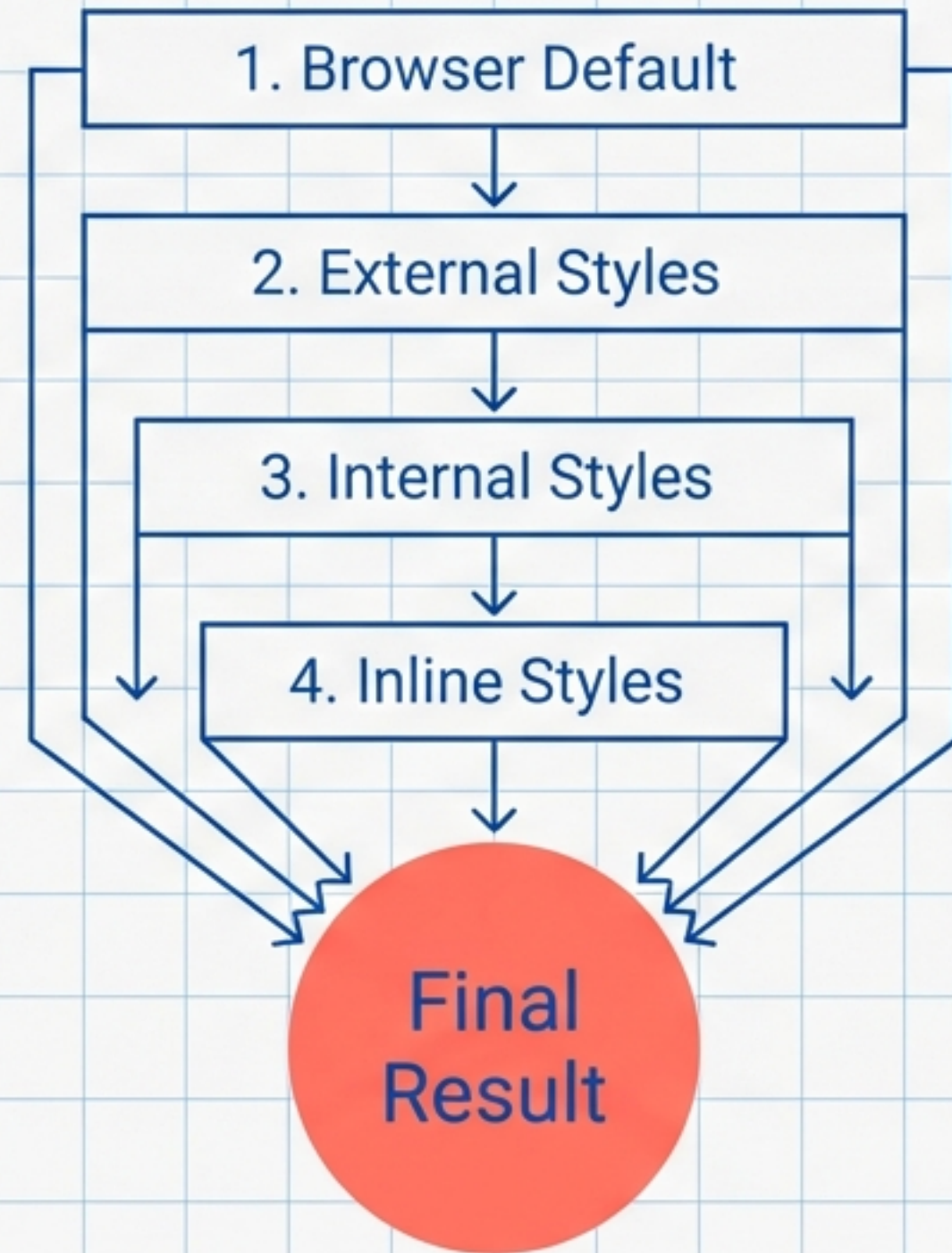
Inline Style

Placed inside the HTML tag. **Warning: Avoid wherever possible.** It mixes content with presentation and breaks the efficiency of CSS.

```
<p style="color: red;">This is inline styled text.</p>
```

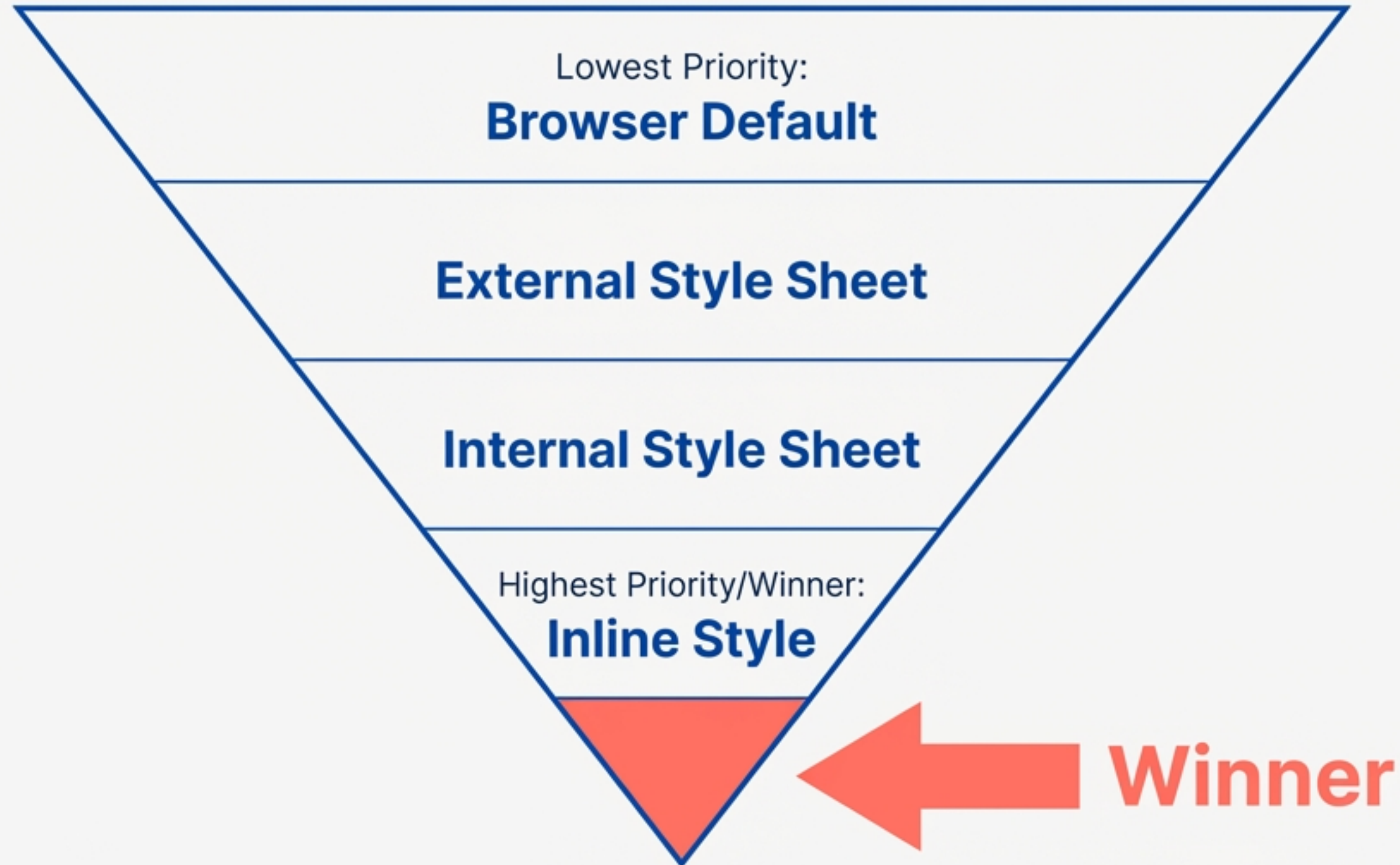

The 'Cascading' Effect

Cascading reflects the way styles are applied to elements. Declarations flow **down** from multiple origins to calculate the final appearance.



The Hierarchy of Power

When styles conflict, the “nearest” style wins.



Precision Targeting: CSS Selectors

Selectors determine exactly which HTML element a rule applies to.



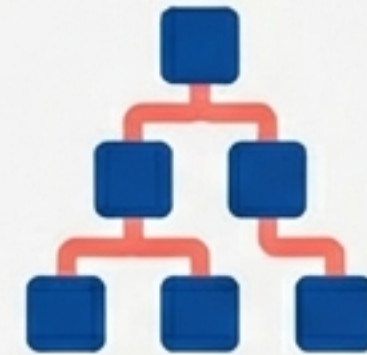
Type Selectors

Target all elements of a specific tag (e.g., all **<h1>** tags).



Attribute Selectors

Target specific IDs or Classes (e.g., **.bold** or **#menu**).



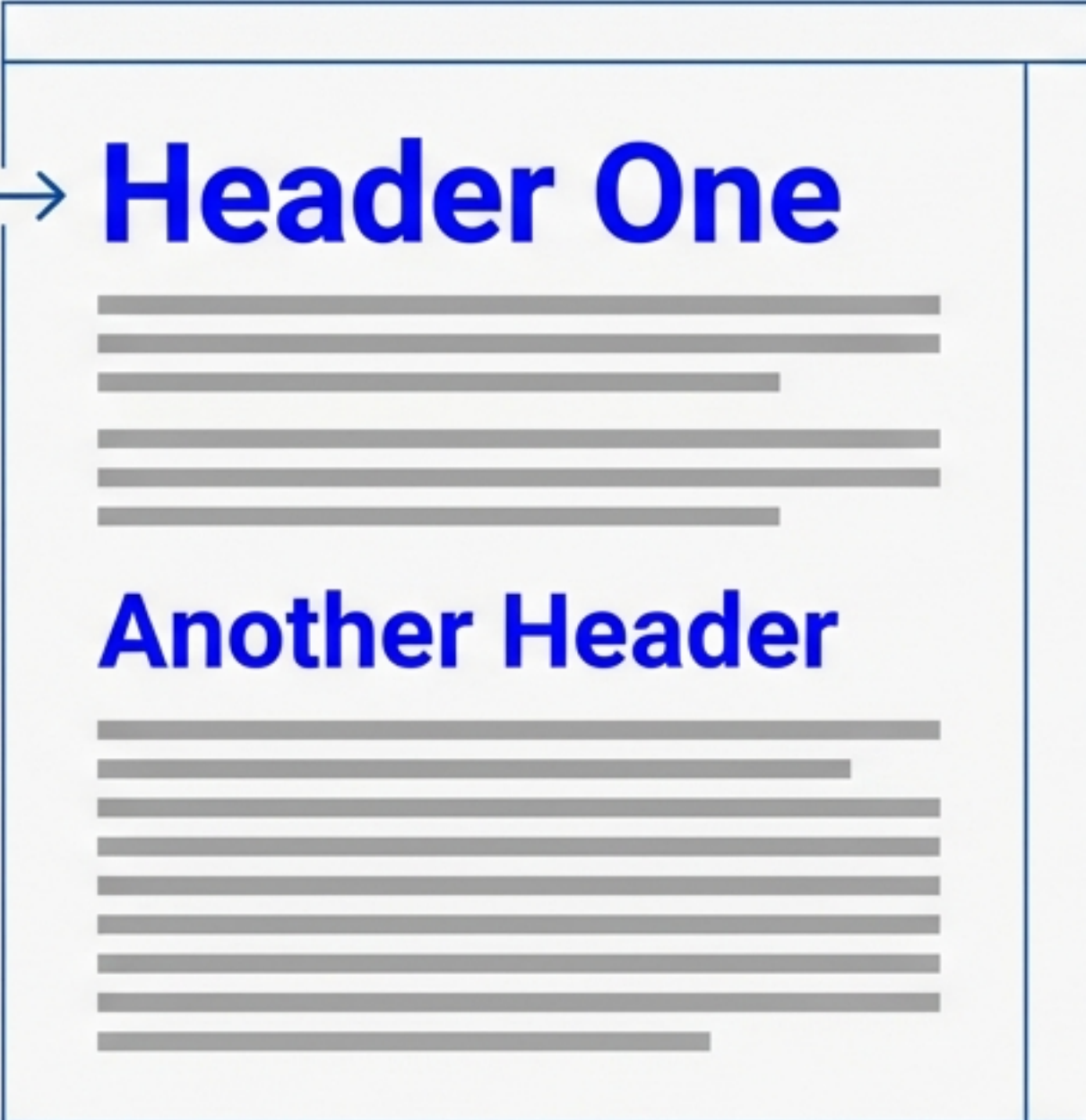
Context Selectors

Target elements based on their nesting or relationship in the document tree.

The Type Selector

```
h1 {  
  color: blue;  
}
```

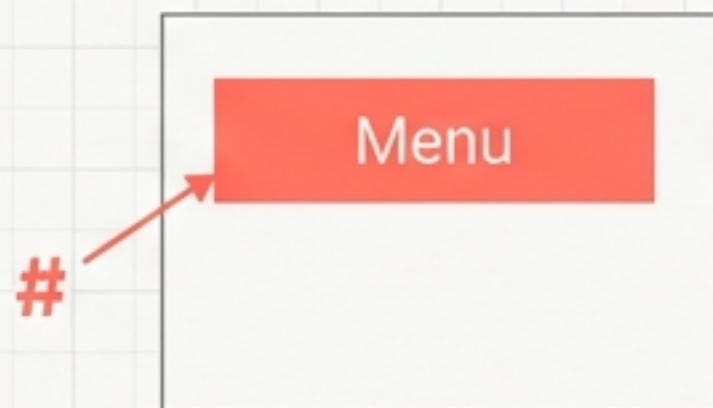
Selects elements based on their HTML tag name (P, H1, BODY, etc.). This acts as a global rule for that tag.



Header One

Another Header

Precision Tools: ID vs. Class



Unique

Defines a style for a **single, unique** element.

```
#menu {  
  padding-top: 8px;  
}
```

Matches that
<div id="menu">.



Reusable

Defines a style applied to **multiple** elements.

```
.bold {  
  font-weight: bold;  
}
```

Matches that
<p class="bold">.

Tip: Elements can hold multiple classes, e.g., <p class='bold large'>.

Summary & Challenge

Summary

- **Separate:** Keep Content (HTML) separate from Presentation (CSS).
- **Link:** Use External stylesheets (`<link>`) for efficiency.
- **Target:** Use Classes (.) for reusable styles and IDs (#) for unique ones.

Coding Challenge

1. Create an HTML document with two paragraphs.
2. Use CSS classes to make one paragraph right-aligned (`p.right`) and one center-aligned (`p.center`).
3. Create a header and apply a unique color to it using an ID.